SPECTRAL ANALYSIS TO STUDY FULL DISCRETIZATIONS OF THE CONVECTION EQUATION

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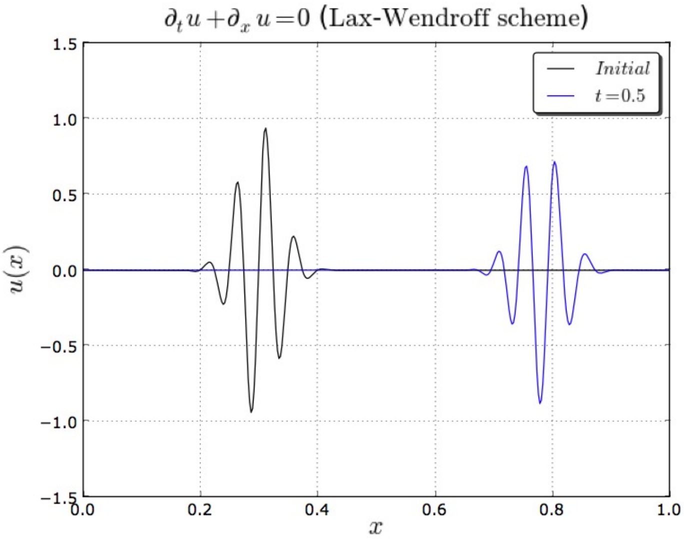

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Spectral analysis to study full discretizations of the convection equation

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Abstract

The purpose of this document is to recall some essential definitions of spectral analysis to evaluate difference approximations of the first-order wave equation. The concepts of dispersion relations and group velocities are first presented. Space and time Fourier analyses are successively described and their benefits are pointed out. Space analysis is available on infinite domains. It determines oscillation evolution, in terms of propagation velocities and amplitude changes, given a certain wavenumber. Time analysis is useful when there are interfaces such as mesh refinement or boundaries, to name a few. The roots of the so-called characteristic equation provide similar information as space Fourier transforms, except it is devoted to pulsations. This allows to introduce the notion of physical and spurious wave solutions. The Lax-Wendroff scheme is used as an illustration throughout this work.

Key words: Convection equation, Spectral analysis, Diffusion / Dispersion properties, Group velocities

Contents

1. Introduction

This document is devoted to the concept of dispersion relations and group velocities to study full discretizations. This kind of study is common for semi-discretizations [2, 11]. In this work, Ω refers to the computational domain and $\Omega \subset \mathbb{R}^d$, with d the number of dimensions. Transport phenomena in fluid are described by solving first-order hyperbolic partial differential equations (PDE). Other wave propagation from other disciplines (acoustics, electromagnetism, neutronics) can also be expressed as systems of first-order hyperbolic PDEs. The main focus is given to the most simple example of such equations: the linear convection equation. Moreover, the classical Lax-Wendroff (LW) scheme is used as the cornerstone of this report.

The linear convection equation reads:

$$\frac{\partial u}{\partial t} + \boldsymbol{a} \cdot \nabla u = 0, \tag{1.1}$$

with $u = u(\boldsymbol{x}, t)$ the unknown, $\boldsymbol{a}$ the (constant) convective velocity. The exact solution of Eq. (1.1) is:

$$u(\boldsymbol{x}, t) = u_0(\boldsymbol{x} - \boldsymbol{a}t),$$

where $u_0(\boldsymbol{x}) = u(\boldsymbol{x}, 0)$ is the initial solution.

In this document, we choose $d = 1$ and $a > 0$ for simplicity. The concepts, presented hereafter, can be generalized to $d > 1$, but this introduces new difficulties like anisotropy, which will be mentioned elsewhere. Besides, the discretization of Ω, noted $\mathcal{T}_h$, is made using regular elements, which leads to the mesh size $h = cst$. We note U_j^n the solution of the discretization of Eq. (1.1), at the discrete time $t^n = nk$, where k is the time step, and associated with node (or control volume) located at $x_j = jh$. It approximates $u(x_j, t^n)$. The LW differencing scheme is written[6]:

$$U_j^{n+1} = U_j^n - \frac{1}{2}c\left(U_{j+1}^n - U_{j-1}^n\right) + \frac{1}{2}c^2\left(U_{j+1}^n - 2U_j^n + U_{j-1}^n\right), \tag{1.2}$$

which can be also written:

$$ZU_j^n = \left\{ I - \frac{c}{2}\left(E - E^{-1}\right) + \frac{1}{2}c^2\left(E - 2I + E^{-1}\right) \right\} U_j^n, \tag{1.3}$$

where we have introduced the identity I, space- E and time-shift Z operators, and:

$$U_j^n = IU_j^n, \qquad U_{j+1}^n = EU_j^n, \qquad U_j^{n+1} = ZU_j^n.$$

The CFL (Courant-Friedrichs-Lewy) number c is used as a parameter to study the stability of the convection schemes and is defined by:

$$c = \frac{ak}{h}.$$

The LW scheme is second-order accurate both in space and time, and is L^2 stable provided $c \leq 1$ [4]. In the present document, we write: $\imath = \sqrt{-1}$.

2. Dispersion relations and definitions

Equation (1.1) admits non-trivial plane-wave solutions of the form:

$$u(x,t) = \exp(\imath\xi x - \imath\omega t)\,, \tag{2.1}$$

provided that the dispersion relation, which links the pulsation ω with the wavenumber ξ,

$$\omega = \xi\, a, \tag{2.2}$$

is respected [7, 10]. Identically, a discrete approximation of Eq. (1.1) admits plane-wave solutions of the form:

$$U_j^n = \exp(\imath\xi jh - \imath\omega nk)\,, \tag{2.3}$$

Therefore, modified dispersion relations can be expressed. Injecting Eq. (2.3) in the difference schemes enables to write an expression, which links ξ and ω.

Example 1. *The dispersion relation of the LW scheme can be obtained from:*

$$\tan(\omega k) = \frac{c\sin(\xi h)}{1 - c^2\left\{1 - \cos(\xi h)\right\}}. \tag{2.4}$$

As explained in [9], as the grid is discrete, the dispersion relations like Eq. (2.4) are multivalued and 2π-periodic in ξh and ωk. In practice and when needed, the relevant branch is chosen by continuity with the low-frequency limit $\omega k \sim c\,\xi h$. From Eq. (2.4), it can be seen with the LW scheme, the dispersion relation related to a numerical scheme is a good approximation of Eq. (2.2) when $\xi h, \omega k \to 0$. On the other hand, it is also a formula involving trigonometric functions. Thus, the relation between ξ and ω is no more linear and the scheme is said to be *dispersive*. Indeed, essential propagation properties will depend on the pulsation or the wavenumber.

The *phase velocity* is the propagation speed of wave crests within a wave and is defined by:

$$v_\varphi(\xi,\omega) = \frac{\omega}{\xi}. \tag{2.5}$$

An even more important notion is the propagation velocity of energy, which is called *group velocity* [7]:

$$\mathcal{V}_g(\xi,\omega) = \frac{d\omega}{d\xi}. \tag{2.6}$$

As Eq. (1.1) is non-dispersive, those two velocities do not depend on ξ or ω and are equal to the convection velocity $v_\varphi = \mathcal{V}_g = a$. Of course, this is not the case of most difference approximations. Dispersion errors imply that wavepacket components will not propagate at the same speed and will be gradually separated, leading to a (sometimes dramatic) shape deformation of the initial signal [10].

It should be mentioned here that the group velocity theory is only valid for *non-dissipative* media or in the limit of *vanishing viscosity*, e.g for $\xi, \omega \in \mathbb{R}$. Any numerical diffusion introduces a frequency-dependent damping of Fourier components. The analysis is then more complicated. The wavepacket amplitude not only varies because of constructive/destructive interferences, but also due to diffusion. Nevertheless, when dispersion dominates, the predictions remain valid, at least for low frequencies [9].

Example 2. *A general formula for the group velocity can be established by differentiating implicitly each side of (2.4) [8, 9, 10]:*

$$k\left(1+\tan^2(\omega k)\right) d\omega = \frac{ch\cos(\xi h)d\xi}{1-c^2\left\{1-\cos(\xi h)\right\}} + \frac{c^3 h\sin^2(\xi h)d\xi}{\left(1-c^2\left\{1-\cos(\xi h)\right\}\right)^2}.$$

Thus:

$$\mathcal{V}_g(\xi,\omega) = \frac{a}{1+\tan^2(\omega k)}\left[\frac{\cos(\xi h)}{1-c^2\left\{1-\cos(\xi h)\right\}} + \frac{c^2\sin^2(\xi h)}{\left(1-c^2\left\{1-\cos(\xi h)\right\}\right)^2}\right]. \qquad (2.7)$$

Clearly, the relation between ξ and ω is non-linear in Eq. (2.7). The interest of a formulae such as (2.4) and (2.7) is also to treat separately the effects of space and time diffencing [9]. In what follows, however, we present other ways to derive dispersion relations and group velocities, which can be viewed as special cases of what has been presented so far. Their interest is that they are more common, rely on very well-known tools or introduce some concepts more easily, in the author's opinion.

The first one is based on *space Fourier analysis* and *the Cauchy problem* (1.1), e.g on an unbounded domain. Given a real wavenumber $\xi \in \mathbb{R}$ (an oscillation in space), one can determine the corresponding amplification factor, in the frame of the *Von Neumann method* [4]. Then, a complex dispersion relation of the form: $\Sigma^\star(\xi) - \imath\Omega^\star(\xi) \in \mathbb{C}$ can be deduced, where the $\star$ denotes the value provided by the numerical scheme. This first strategy is an efficient and classical tool in numerical analysis. It is often enough to get a good insight into a discretization. Nevertheless, for problems involving *mesh refinement* or *bounded domains*, a second analysis based on *time Fourier transforms* has to be used and enables to introduce even more concepts [11]. Another complex dispersion relation can also be written $M^\star(\omega) - iK^\star(\omega) \in \mathbb{C}$ in this case, with $\omega \in \mathbb{R}$.

3. Space Fourier transforms

We now study the main characteristics of numerical schemes, within the frame of space Fourier transforms. We suppose the domain Ω infinite, thus without boundaries. The problem associated with Eq. (1.1) and the initial condition u_0 is a Cauchy problem.

3.1. *Phase errors and propagation velocities*

Let us suppose that U^n is a function with a finite norm, so that it can be expressed using Fourier transforms: Let us suppose that the grid function $\{U_j^n\}_{j\in\mathbb{Z}}$ belongs to $\ell^2(\mathbb{Z})$, so that it admits the Fourier representation:

$$U^n = \int_{-\infty}^{\infty} \widehat{U}^n(\xi) e^{\imath\xi x} \frac{d\xi}{2\pi},$$

with ξ the wavenumber and $\imath = \sqrt{-1}$. Then, applying a linear finite difference scheme leads, after some manipulations[1], to:

$$U_j^{n+1} = \int_{-\infty}^{\infty} \widehat{U}^n(\xi) g(\xi) e^{\imath\xi x_j} \frac{d\xi}{2\pi}.$$

where:

$$g(\xi) = \frac{\widehat{U}^{n+1}(\xi)}{\widehat{U}^n(\xi)}$$

is the *amplification factor* with the wavenumber ξ. On the other hand, the space Fourier transform of Eq. (1.1) is:

$$\frac{\partial \widehat{u}}{\partial t} + \imath\xi a\hat{u} = 0, \tag{3.1}$$

and $\widehat{u}(t^{n+1})/\widehat{u}(t^n) = \exp(-\imath\xi ak)$. As the dispersion relation of Eq. (1.1) is $\omega = \Omega(\xi) = \xi a$, then the expression of the ideal amplification factor is:

$$g^{ex}(\xi) = e^{-\imath\omega k} \tag{3.2}$$

Example 3. *The amplification factor of the LW scheme (1.2) is:*

$$g^{LW}(\xi) = 1 - \imath\, c \sin(\xi h) - c^2 \{1 - \cos(\xi h)\}. \tag{3.3}$$

As can be seen in Eq. (3.3), when a difference formula is used to approximate Eq. (1.1), the pulsation of a harmonic ξ is only an approximation of the real one. Moreover, because in the

[1] Note that E is a Toeplitz operator and $\hat{E} = e^{\imath\xi h}$ [11]

general case, the difference approximation introduces spurious diffusion. Then, the amplicafication factor of a numerical scheme can be written:

$$g(\xi) = e^{s^\star k} = e^{\sigma^\star k - \imath \omega^\star k}. \tag{3.4}$$

Equation (3.4) then allows to write the functional:

$$\begin{array}{rcl} S^\star : \mathbb{R} & \longrightarrow & \mathbb{C} \\ \xi & \longmapsto & S^\star(\xi) = \Sigma^\star(\xi) - \imath \Omega^\star(\xi) = \sigma^\star - \imath \omega^\star, \end{array}$$

$$S^\star(\xi) = \sigma^\star(\xi) - i\,\omega^\star(\xi).$$

which definies the *modified dispersion relation* of the difference scheme. $\omega^\star$ is to be compared with the real pulsation and holds the *phase errors*, while $\sigma^\star$ is linked with the *numerical diffusion* of the scheme. We can also define the modified wavenumber [4]:

$$\xi^\star(\xi) = \frac{\Omega^\star(\xi)}{a}, \tag{3.5}$$

and the modified phase velocity:

$$v_\varphi^\star(\xi) = \frac{\Omega^\star(\xi)}{\xi}. \tag{3.6}$$

Furthermore, a cut-off wavenumber can be defined as the maximum wavenumber, which can be represented by the discretization[2] and is often far smaller than π/h:

$$\xi_c h = \max_{\xi h} \left(\xi^\star h\right).$$

Example 4. *The modified dispersion relation of the LW scheme gives:*

$$\omega^\star = \frac{1}{k} \arctan\left(\frac{c \sin(\xi h)}{1 - c^2\left\{1 - \cos(\xi h)\right\}}\right). \tag{3.7}$$

Then, its modified wavenumber is:

$$\xi^\star h = \frac{1}{c} \arctan\left[\frac{c \sin(\xi h)}{1 - c^2\left\{1 - \cos(\xi h)\right\}}\right]. \tag{3.8}$$

Figure (1) shows the modified wavenumber of the LW scheme for different CFL numbers. The LW scheme has a good phase accuracy for longer wavelengths, but shows a poor behavior for badly resolved scales, especially for small CFL numbers. It is noteworthy that the LW scheme has no phase error[3] at $c = 1$.

[2] The phase error at $\xi = \xi_c$ is often big (up to 50%). Another definition must be written to better evaluate phase accuracy. See [1].

[3] No error at all in fact !

Table (1) gathers the cut-off wavenumbers (and the corresponding wavelengths $\lambda = 2\pi/\xi$) of the LW scheme, depending on the CFL number. Moreover, wavenumbers $k_a h$, such that:

$$|\xi^\star(\xi_a) - \xi_a| / \xi_a < 0.01$$

are also added in Tab. (1), to assess the phase accuracy.

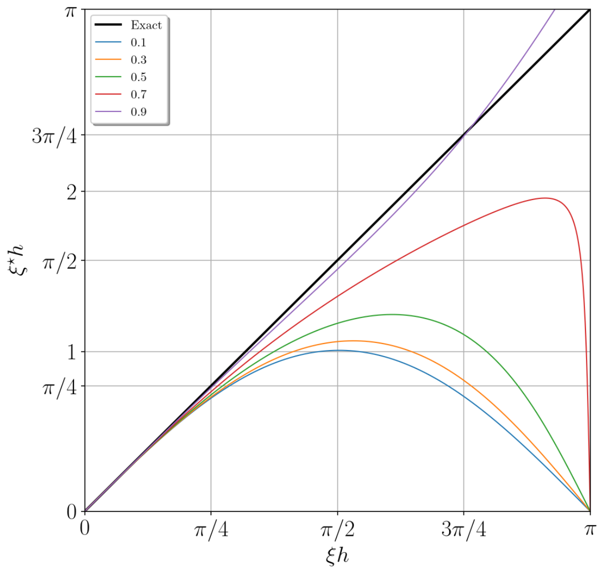

Figure 1: Modified wavenumber of the LW scheme for CFL from 0.1 to 0.9.

c	$\xi_c h$	λ_c/h	$\xi_a h$	λ_a/h
0.1	1.007	6.24	0.245	25.6
0.3	1.066	5.89	0.251	25.0
0.5	1.231	5.10	0.283	22.2
0.7	1.959	3.21	0.346	18.2
0.9	*3.491*	*1.80*	0.591	10.6

Table 1: $\xi_c h$ and $\xi_a h$ for the LW scheme and different CFL numbers.

Table (1) confirms that phase errors are lower when $c \to 1$. It also shows that $\xi^\star$ is relatively small, compared with π (the supposedly maximum non-dimensionalized wavenumber), which is linked with the strong discrepencies of the scheme with small wavelengths. Phase errors are bigger than 1%, as soon as the wavelength is smaller than around $20h$, for a large band of CFL numbers ($c < 0.5$). This shows that dispersion errors are non-negligible, even when resolution seems rather correct (15-20 nodes per wavelength).

As said in section 2, group velocity can be used to further describe wave packets and energy transport. Knowing the modified pulsation, the group velocity of the numerical scheme can be calculated in the same way as the exact one:

$$\mathcal{V}_g^\star(\xi) = \frac{d\omega^\star}{d\xi} = a\frac{d\xi^\star}{d\xi}. \tag{3.9}$$

Depending on the resolution power of the difference schemes, some waves can have an unexpected behavior. Group velocity is the main tool to discriminate those waves, as it gives information about the way they propagate. The longer wavelenghts ($\xi h \to 0$) behave like physical waves (consistency condition) and moves in the right direction, at the right velocity and dispersion errors are relatively small, especially when a high-order scheme is used. On the other hand, the smaller wavelengths ($\xi h \to \pi$), which are badly resolved (only few points), are moving at a wrong speed, in the opposite direction. They can be considered as spurious waves, as they do not follow the physical laws and can threaten the stability of the computation. Several illustrations are provided in [9, 11]. Vichnevetsky and Bowles [11] have called the first oscillations *p-waves* and the spurious ones *q-waves*.

A good criterion to distinguish waves is thus the sign of the group velocity:

Definition 1. *If the product a $\mathcal{V}_g^\star < 0$, then the corresponding wave ξ is a spurious one, which is called q-wave.*

q-waves can be very dangerous, as they carry some energy in the wrong direction or be stationary, can reflect on boundaries and eventually lead to numerical instabilities [9, 11].

Example 5. *The modified group velocity of the LW scheme is:*

$$\frac{\mathcal{V}_g^\star}{a} = \frac{(1-c^2)\cos(\xi h) + c^2}{c^2\sin^2(\xi h) + \{c^2\cos(\xi h) - c^2 + 1\}^2}, \tag{3.10}$$

and Figure (2) displays it for different CFL numbers. As explained above, the scheme behaves rather well for $\xi h \to 0$, especially for higher CFL numbers. In general, phase errors are very significant when $\lambda \approx 10 - 20h$. When oscillations are smaller than around $4h$, they are even convected in the wrong direction (especially when the CFL number is small). One should also note that there are no q-waves at higher CFL numbers (see $c = 0.9$).

3.2. *Diffusion errors*

So far, we have only considered phase errors, but most numerical schemes also introduce diffusion errors, as soon as $\Sigma^\star$ is not zero. Indeed, we have:

$$|g(\xi)| = \exp(\sigma^\star k). \tag{3.11}$$

Consequently:

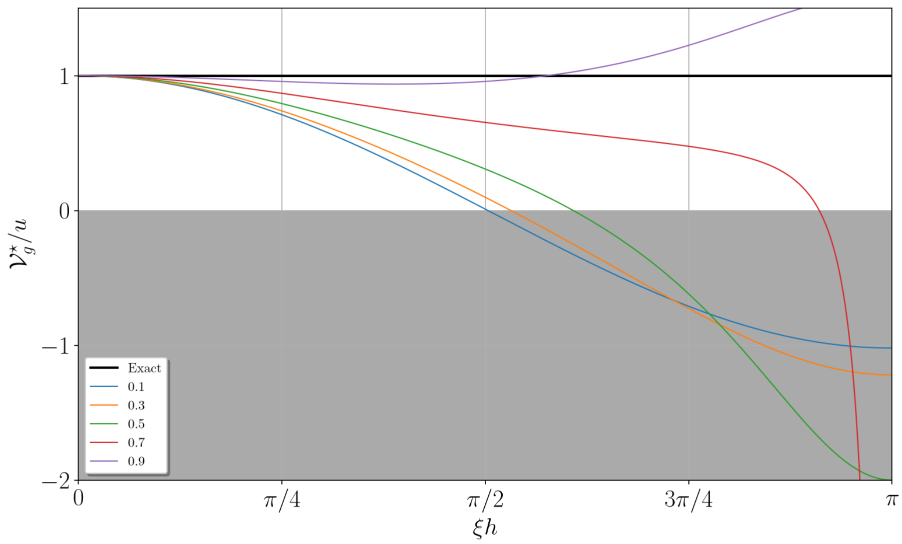

Figure 2: Modified group velocity of the LW scheme for CFL from 0.1 to 0.9. The shaded area corresponds to negative group velocity waves (spurious waves).

- if $\exists\, \xi$ such that $\sigma^\star > 0$, the method is unstable (exponential growth),
- if $\sigma^\star = 0$, the method is non-dissipative or marginally stable,
- if $\exists\, \xi$ such that $\sigma^\star < 0$, the method is dissipative.

Numerical diffusion implies a decay of wave amplitudes and a smearing of strong gradients. This phenomenon is usually not desirable, for example to simulate turbulence. Nevertheless, it tends to stabilize a numerical method and often eliminates spurious waves, generated by phase errors, the Gibbs phenomenon, or spurious reflections at interfaces (mesh refinement, boundaries...). However, we recall that the group velocity theory only works in the limit of zero or vanishing viscosity [9]. Indeed, the amplitude decay is due to two phenomena: diffusion and destructive interferences due to dispersion. Therefore, the amplification factor should just be used to study numerical diffusion of sine waves.

Example 6. *Figure (3) displays the amplification factor modulus of the LW scheme for different CFL numbers. For small CFL numbers, the LW is hardly dissipative, even for high wavenumbers. As* $c \to \sqrt{2}/2$, *numerical diffusion gets stronger, especially for* $\xi h \to \pi$. *Then, as* $c \to 1$, *the scheme is again less dissipative. It is stable for* $c \leq 1$. *As an example, Fig. (3) shows* $|g(\xi)|$ *for* $c = 1.1$, *which belongs to the unstable region and leads to the simulation blow-up.*

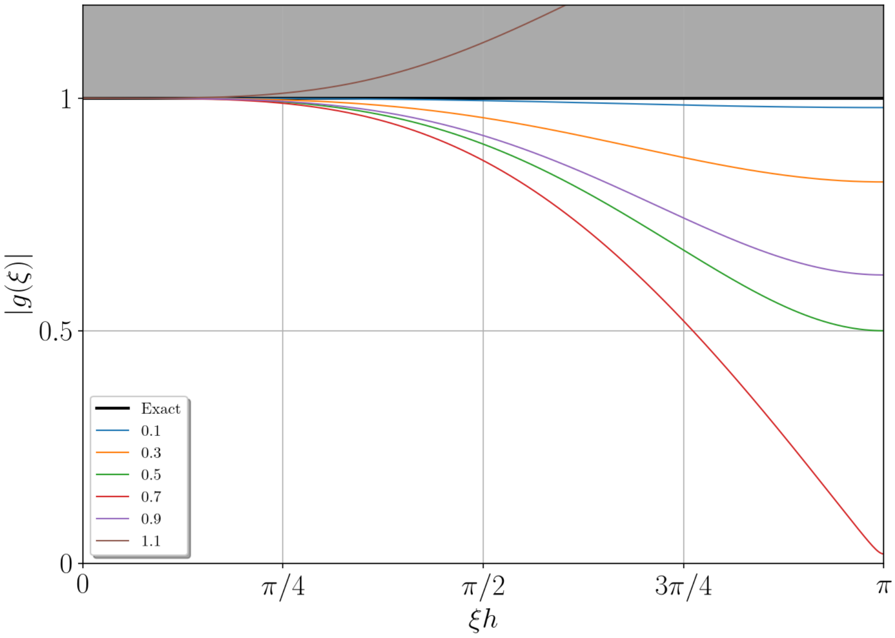

Figure 3: Amplification factor modulus of the LW scheme for CFL from 0.1 to 1.1. The shaded area corresponds to the unstable region.

4. Time Fourier transforms

We have shown that space Fourier transforms are very useful to study wave propagation within the frame of the Cauchy problem (initial value problem with infinite domains). Despite the power of this tool, it remains difficult to analyze what happens next to boundaries or interfaces. To fix this problem, Vichnevetsky proposes to work with time Fourier transforms [11].

4.1. Characteristic equation and dispersion relations

Let suppose that U_j is a function with a finite norm, so that it can be expressed using (discrete) time Fourier transforms:

$$U_j^n = \int_{-\pi/k}^{\pi/k} \breve{U}_j(\omega) e^{\imath\omega n k} \frac{d\omega}{2\pi},$$

and:

$$\breve{U}_j(\omega) = k \sum_{-\infty}^{+\infty} U_j^n e^{-\imath\omega n k}.$$

Following the methodology of [11] and using the time Fourier transform[4], the *characteristic equation* of a difference scheme can be obtained:

$$f(c, \omega k, \breve{E}, \breve{E}^2, \breve{Z}, \dots) = 0.$$

[4]It is worth recalling $\breve{Z} = e^{\imath\omega k}$.

The characteristic equation has generally *several roots* $\check{E}_\alpha$ and the time Fourier transform of the solution is composed with several components:

$$\check{U}_j(\omega) = \sum_\alpha \check{\phi}_{j,\alpha}(\omega) = \sum_\alpha \check{\phi}_{0,\alpha}\left[\check{E}_\alpha(\omega)\right]^j,$$

Using time Fourier transforms, Eq. (1.1) can also be rewritten:

$$\imath\omega\check{u} + a\frac{\partial\check{u}}{\partial x} = 0, \tag{4.1}$$

Then $\check{u}(x_{j+1})/\check{u}(x_j) = \exp(-\imath\omega x/a)$. As the dispersion relation of Eq. (1.1) is $\xi = K(\omega) = \omega/a$, then the exact space-shift operator[5] is $\check{E}^{ex} = \exp(-\imath\xi h)$.

Similarly to the work done with space Fourier transforms, we can introduce *modified pulsation, wavenumber, phase and group velocities*, writting:

$$\check{E}(\omega) = e^{\tau^\star h} = e^{\mu^\star h - \imath\,\xi^\star h}. \tag{4.2}$$

Eq. (4.2) allows to write a functional:

$$\begin{array}{rcl} K^\star : \mathbb{R} & \longrightarrow & \mathbb{C} \\ \omega & \longmapsto & T^\star(\omega) = M^\star(\omega) - iK^\star(\omega) = \mu^\star - \imath\xi^\star, . \end{array}$$

which is the dispersion relation of the root $\check{E}$. By analogy with section 3, $\xi^\star$ is to be compared with the real wavenumber and holds *the phase errors* of the scheme, while $\Im(\xi^\star)$ is linked with the numerical diffusion. Therefore, the real part of $K^\star$ is the modified dispersion relation.

The modified dispersion relation also leads to the definitions:

$$\omega^\star(\omega) = K^\star(\omega)a = -\frac{a}{h}\arctan\left(\frac{\Im(\check{E})}{\Re(\check{E})}\right) \tag{4.3}$$

$$v_\varphi^\star(\omega) = \omega/K^\star(\omega) = -\frac{a\omega k}{c}\left\{\arctan\left(\frac{\Im(\check{E})}{\Re(\check{E})}\right)\right\}^{-1} \tag{4.4}$$

$$\mathcal{V}_g^\star(\omega) = \frac{a}{(\partial\omega^\star/\partial\omega)}. \tag{4.5}$$

From relation (4.3), one can also calculate wavelengths associated with a pulsation ω:

$$\lambda^\star(\omega) = \frac{2\pi}{K^\star(\omega)}.$$

Example 7. *Using the methodology described above, the LW scheme (1.2) becomes:*

$$\check{Z} = 1 - \frac{c}{2}\left(\check{E} - \check{E}^{-1}\right) + \frac{c^2}{2}\left(\check{E} - 2 + \check{E}^{-1}\right).$$

[5] Sometimes called *space* amplification factor.

This leads to the characteristic equation of the LW scheme:

$$\breve{E}^2 + \underbrace{\frac{2\left(1 - c^2 - \breve{Z}\right)}{c(c-1)}}_{b(\omega)} \breve{E} + \frac{c+1}{c-1} = 0. \tag{4.6}$$

The roots of Eq. (4.6) are:

$$\breve{E}_{p/q} = \frac{-b(\omega) \pm \sqrt{\Delta(\omega)}}{2},$$

with:

$$\Delta(\omega) = b(\omega)^2 - 4\,\frac{c+1}{c-1}.$$

It means there is two kind of waves, which can propagate and the time Fourier transform of the solution is:

$$\breve{u}_j(\omega) = \breve{p}_j(\omega) + \breve{q}_j(\omega),$$

such that:

$$\begin{array}{rcl} \breve{p}_j & = & \breve{p}_0 \left[\breve{E}_p\right]^j \\ \breve{q}_j & = & \breve{q}_0 \left[\breve{E}_s\right]^j \end{array}$$

In comparison with the space Fourier transforms, the main advantage is to highlight that there exists several waves associated with the scheme. For instance, in the case of the LW scheme, there are two waves for a given pulsation ω. This is due to the fact that a three-point explicit scheme has a quadratic characteristic equation. The first one, corresponding to the well-resolved wavenumbers, are named *p-waves* or physical / smooth waves. Their group velocity is such that $a\,\mathcal{V}_g^\star > 0$. On the other hand, there are also badly-resolved harmonics and can be called *q-waves* or spurious / saw-tooth waves. Their group velocity is such that $a\,\mathcal{V}_g^\star < 0$. Thus, they propagate upstream, which is wrong and explains their name. As in the case of space analysis, we can introduce the definitions:

Definition 2. *Considering a root $\breve{E}$, when $\omega k \to 0$:*

- *if $a\,\mathcal{V}_g^\star > 0$, then the root is a p-wave.*
- *if $a\,\mathcal{V}_g^\star < 0$, then the root is a q-wave.*

This definition is to be carefully used when the difference approximation is dissipative, as it is discussed in paragraph 4.3.

Example 8. *Figures (4), (5) and (6) display an example of modified pulsation, wavelenghts and group velocities for the LW scheme for a CFL number of $c = 0.1$. Again, there is two solutions for a given pulsation. As far as the p-wave is concerned, the LW scheme has a satisfactory behavior for smaller pulsations. $\omega^\star$ is close to ω, waves are well-represented ($\lambda^\star \approx \lambda$) and propagate at the right velocity ($\mathcal{V}_g^\star/a \to 1$). At the same time, the q-wave, badly-resolved (close to $2h$), also exists for the same pulsations and propagate in the opposite direction. As pulsation goes bigger, phase errors are more sensible and the behavior of both waves gets closer.*

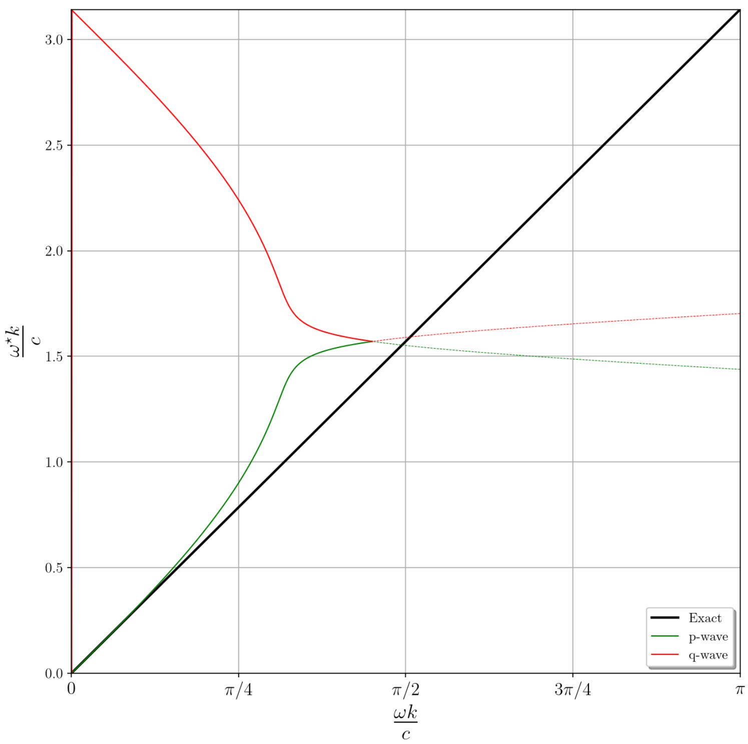

Figure 4: Modified pulsations of the LW scheme depending on the pulsation ω. Black line: exact. Green line: *p-waves*. Red line: *q-waves*. CFL number is $c = 0.1$. Dashed lines are, in this case, evanescent solutions.

4.2. Diffusion errors

As mentionned in section 3, most numerical schemes introduce diffusion errors. Indeed, as soon as $\mu^\star$ is not zero, the corresponding root is diffused or anti-diffused. Suppose that $a > 0$. If $u\ \mathcal{V}_g^\star$ is positive, this is a physical wave and:

- $\mu^\star = 0$, the scheme is non-dissipative.
- if $\exists\ \omega$ such that $\mu^\star > 0$, the scheme is unstable.
- if $\exists\ \omega$ such that $\mu^\star < 0$, the scheme is dissipative.

Inversely, if $a\ \mathcal{V}_g^\star$ is negative, this is a spurious wave. Then it moves in the wrong direction and:

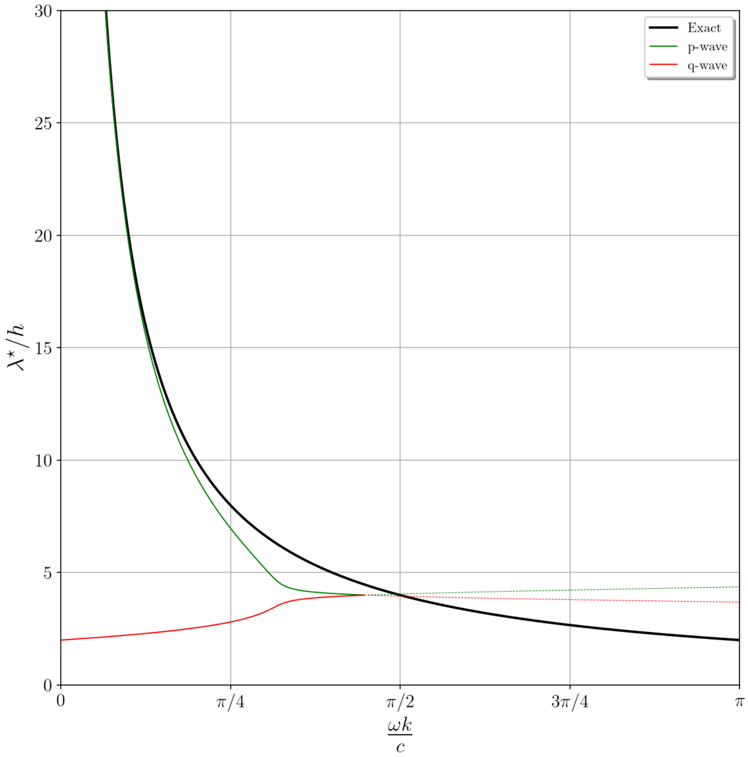

Figure 5: Modified wavelenghts of the LW scheme depending on the pulsation ω. Black line: exact. Green line: *p-waves*. Red line: *q-waves*. CFL number is $c = 0.1$. Dashed lines are, in this case, evanescent solutions.

- $\mu^\star = 0 = 0$, the scheme is non-dissipative.
- if $\exists\, \omega$ such that $\mu^\star = 0 < 0$, the scheme is unstable.
- if $\exists\, \omega$ such that $\mu^\star = 0 > 0$, the scheme is dissipative.

Example 9. *Figure (7) shows the modulus of the LW scheme characteristic equation roots $\breve{E}_{p/s}$ for a CFL number $c = 0.1$. Using Fig. (5), we can see that the smooth waves are slowly diffused, until $\lambda \approx 5 - 10h$. On the other hand, all the spurious waves (red line) are diffused at this CFL number. Beyond a cut-off frequency $\omega k/c \approx 1.4$, solutions are very rapidly dissipated and are refered as evanescent solutions (dashed lines on the figures).*

4.3. Evanescent solutions and cut-off frequency

We finally give a brief discussion about evanescent solutions and the definition of *p-* and *q-waves*. Using LW as the reference, we only consider difference scheme with a quadratic characteristic equation, but this can be generalized for higher-order equations.

In the examples given above (see Figs. (4)-(6)), some part of the solutions are indicated with dashed lines. Going from a continuous to a dash line corresponds to a rapid change of behavior,

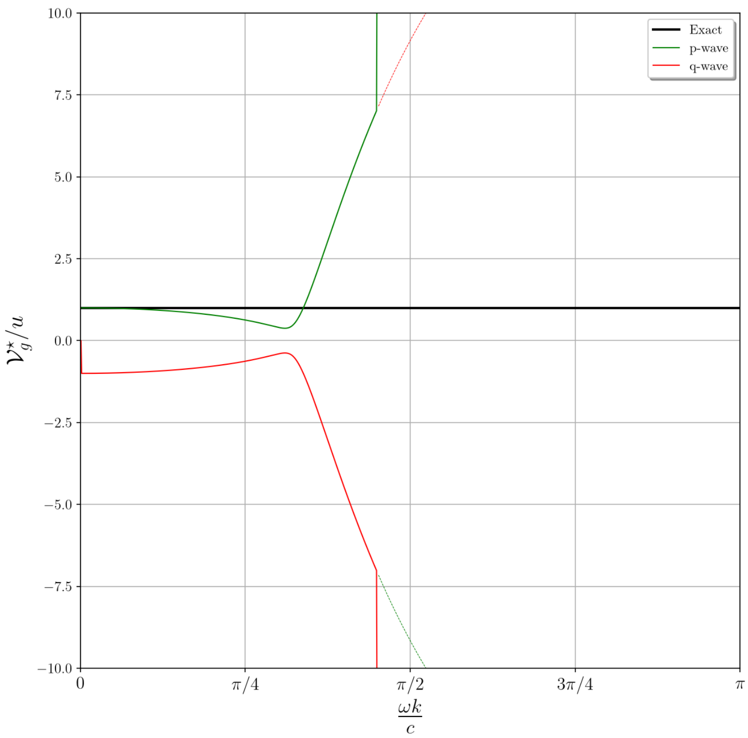

Figure 6: Modified group velocities of the LW scheme depending on the pulsation ω. Black line: exact. Green line: *p-waves*. Red line: *q-waves*. CFL number is $c = 0.1$. Evanescent solutions do not appear on screen.

as $\omega k/c$ grows. The sign of group velocity changes, $|E|$ goes from a value bigger (*resp.* smaller) to a value smaller (*resp.* bigger) than 1. This change occurs at the same value of $\omega k/c$ for both roots[6]. The dashed part of the roots of Eq. (4.6) are strongly damped for $c = 0.1$. It is then straightforward to define a cut-off frequency $\omega \mid \omega_p^\star = \omega_q^\star$. Pulsating at a frequency beyond this limit results in no wave in the domain. This phenomenon is even clearer with a non-dssipative scheme, as highlighted by Vichnevetsky and Bowles [11], for the second-order centred scheme. In such a case, solutions are evanescent beyond the cuf-off frequency. Those similar behaviors are not surprising as the LW scheme is hardly dissipative and close to the second-order centred scheme, for $c \to 0$.

For higher CFL numbers, conclusions are somewhat different, in the case of diffusive schemes, like LW. They remain roughly the same for *p-waves* (*a* $\mathcal{V}_g^\star > 0$ and $|E_p| \to 1^-$ for $\omega k/c \to 0$). However, *q-waves* can become evanescent when $\omega k/c \to 0$ and *a* $\mathcal{V}_g^\star < 0$ and, on the other hand, scarcely damped as $\omega k/c \to \pi$ ($|E_q| \to 1^-$ and *a* $\mathcal{V}_g^\star > 0$). As a consequence, all the non-evanescent waves are convected in the right direction and whatever the pulsation, there is a corresponding wave and therefore, no cut-off frequency. Therefore, definition 2 seems to be no more valid, as a root designed as a *q-wave*, thus a supposedly spurious wave, can behave correctly, hence rather as a *p-wave*, for high pulsations.

[6]In the case of a quadratic characteristic equation.

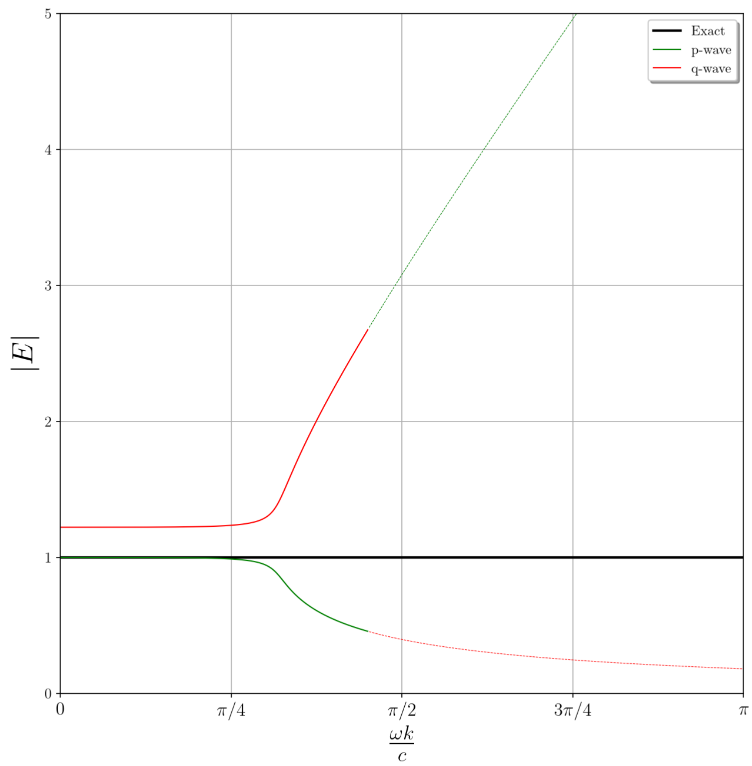

Figure 7: Roots modulus of the LW scheme characteristic equation, depending on the pulsation ω. Black line: exact. Green line: *p-waves*. Red line: *q-waves*. CFL number is $c = 0.1$. Dashed lines are evanescent solutions.

5. Numerical experiences

Some examples, once again based on the Lax-Wendroff scheme, give a demonstration of the power of space and time spectral analyses in this section.

5.1. Space analysis

In this first paragraph, we show the evolution of two wavepackets and a sine wave. The domain $\Omega = (0, 1)$ is periodic and meshed with $N_x = 100$ points, $a = 1$ and $k = 10^{-3}$, so that $c = 0.1$. The initial wavepackets are centred around $x = 0.5$.

- **Wavepacket A:** The wavepacket has a carrier wave, which wavenumber is $\xi h = \pi/5$. After 1000 iterations ($t = 1$), the packet is ideally supposed to superimpose with the initial solution.
- **Wavepacket B:** The wavepacket has a carrier wave, which wavenumber is $\xi h = \pi$. After 100 iterations ($t = 0.1$), the packet should be centred around $x = 0.6$.

- **Signal C:** We also show that space analysis predicts amplitude errors, as well. As the study of wavepackets is not stricly suitable to evaluate dissipative effects [7], we use a sinusoidal periodic signal with a wavenumber $\xi h = \pi/10$.

Case A: Figure (8) shows the wavepacket after, supposedly, 1 turn ($t = 1$). Actually, as the associated group velocity is less than 1, it is not superimposed with the initial solution. There is a space shift and the wavepacket is centred around $x = 0.3$, which means its propagation velocity has been around 0.8. This value is consistant with theory, as Eq. (3.10) gives the group velocity: $\mathcal{V}_g^\star(\pi/5) \approx 0.81$ (with a CFL number of 0.1).

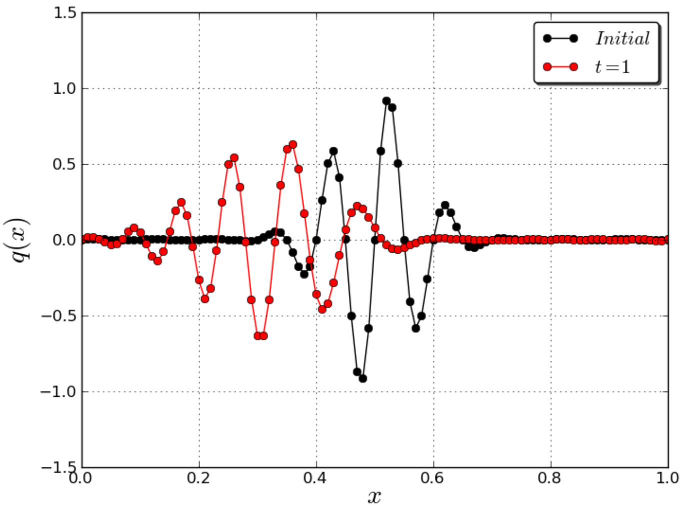

Figure 8: Convection of wavepacket A with the LW scheme, after 1000 iterations. Black line: Initial solution. Red line: Numerical solution. CFL number is $c = 0.1$.

Case B: Figure (9) shows the wavepacket after, supposedly, 0.1 turn ($t = 0.1$). It can be seen that the wavepacket has travelled in the opposite direction and is centred around $x = 0.4$. Its propagation velocity has thus been around -1. Again, this value is predicted by theory, as Eq. (3.10) gives the group velocity: $\mathcal{V}_g^\star(\pi) \approx -1$ (with a CFL number of 0.1).

Case C: Figure (10) shows the sine wave after, supposedly, 10 turns ($t = 10$). It can be seen that the wave suffers from amplitude and phase errors, but in the present case, we are just interested in the former. The wave amplitude is a bit less than 0.9. From Eq. (3.3), the amplification factor modulus is $|g(\pi/10)| \approx 0.999988$. Then, after 10000 iterations, the initial wave amplitude is multiplied by 0.888, which corresponds the numerical results.

This couple of illustrations have shown the interest of spectral analysis in space. It enables to predict the resolution power of a scheme and discriminate physical and spurious waves,

[7] As it has been recalled in sections 2 and 3, even though the group velocity theory works very well in the limit of vanishing viscosity [5], it is only valid, stricly speaking, when there is no dissipation. The dispersion effects make it hard to see if an amplitude decrease is due to diffusion or destructive interferences.

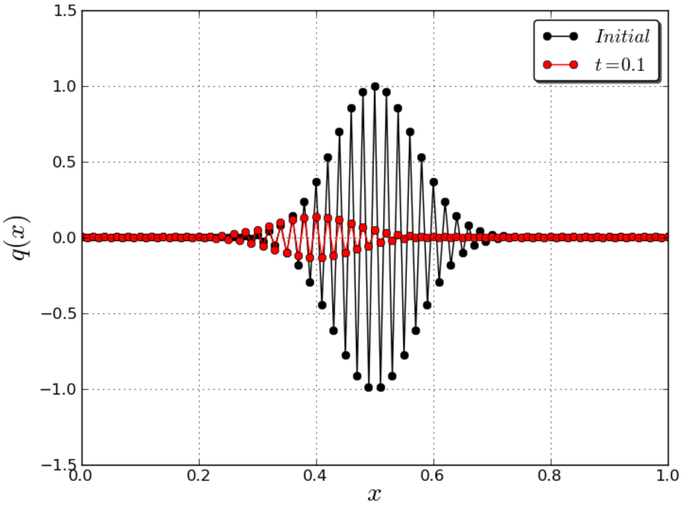

Figure 9: Convection of wavepacket B with the LW scheme, after 100 iterations. Black line: Initial solution. Red line: Numerical solution. CFL number is $c = 0.1$.

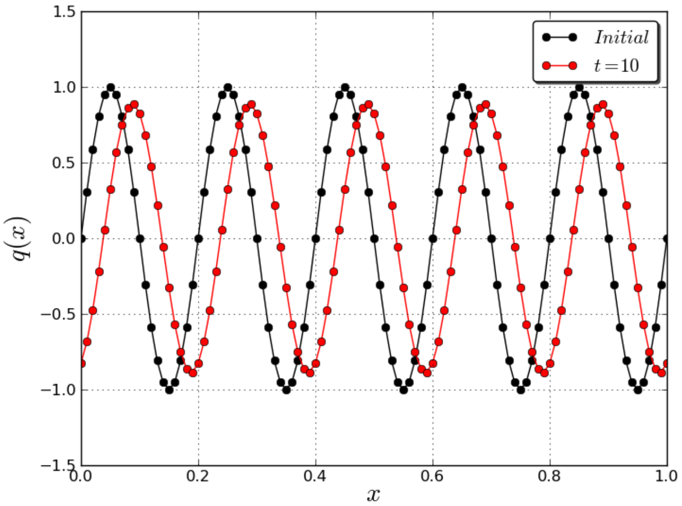

Figure 10: Convection of a sine wave with the LW scheme, after 10000 iterations. Black line: Initial solution. Red line: Numerical solution. CFL number is $c = 0.1$.

depending on phase and group velocities. Moreover, the amount of numerical diffusion introduced by the discretization can also be calculated and L^2 stability parameters can be determined. Knowing those main characteristics allows to anticipate many errors when using a numerical scheme. Furtherfore, stabilization or filter operators can be designed to destroy those spurious waves, as it can be seen that their behaviour is non-physical, and often dangerous.

5.2. *Time analysis*

Similarly to the work of [9], we introduce a source term at $x = 0.5$ to illustrate the use of time Fourier transforms. We study two kind of source terms:

- **Smooth source term D:** The domain is composed with $N_x = 200$ knots, the CFL is still $c = 0.1$. The source term is defined such that $u(0.5, t) = \sin(20\pi t)$, which is $\omega k/c = \pi/10$. After 200 iterations ($t = 0.1$), we are supposed to obtain one period of a sine wave, which front has reached $x = 0.6$.

- **Saw-tooth source term E:** The domain is composed with $N_x = 1000$ nodes and is still periodic but also bigger: $\Omega = (0, 4)$. The CFL is still $c = 0.1$. The source term is such that $u(2, t) = (-1)^n$, then $\omega k/c = 10\pi$. Thus, it is far greater than the cut-off pulsation. The result is given after 2500 iterations ($t = 1$).

Case D: Figure (11) shows the results obtained after 200 iterations, which is one period in time for the source term. It can be seen that there is one period of a smooth sine wave at $x > 0.5$, which has travelled at $\mathcal{V}_g^\star \approx 1$, as expected by theory. On the other hand, there is also some spurious oscillations at $x < 0$, which carrier wavelength is around $2h$. They have travelled at $\mathcal{V}_g^\star \approx -1$, as predicted by theory (see Figs. (4), (5) and (6)). One should note that the *q-waves* are also strongly dissipated, as it can been shown on Fig. (7).

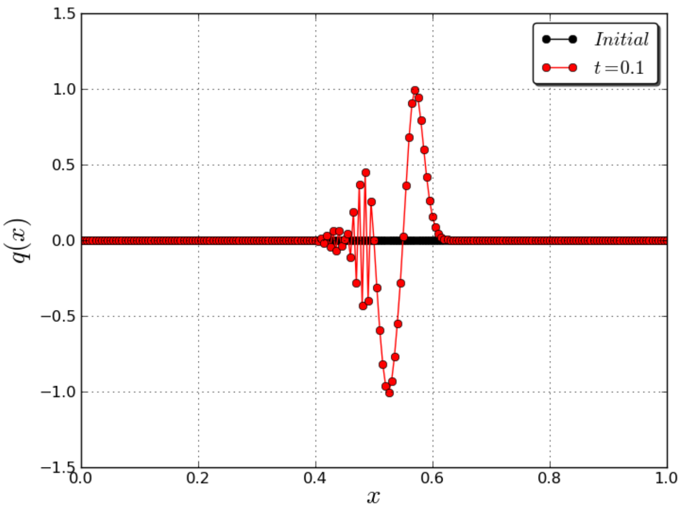

Figure 11: Source term D at $x = 0.5$. Result after 200 iterations. Black line: Initial solution. Red line: Numerical solution. The scheme is LW and the CFL number is $c = 0.1$.

Case E: Figure (12) shows the results obtained after 2500 iterations. Voluntarily, the pulsation was chosen far greater than the cut-off pulsation. As theory predicted, nothing can be seen (evanescent waves), except a transient phenomena (see the oscillations for $x \in (2.5, 3)$) and the source term itself at $x = 2$. If a zoom is made near the source terms, very small and strongly damped oscillations can be seen each side of the source term, with wavelengths close to $4h$.

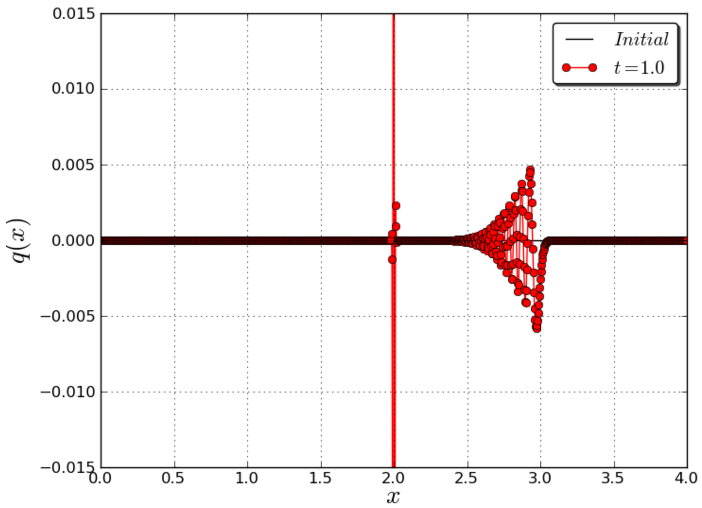

Figure 12: Source term E at $x = 2$. Result after 2500 iterations. Black line: Initial solution. Red line: Numerical solution. The scheme is LW and the CFL number is $c = 0.1$.

Those examples have confirmed the interest of spectral analysis in time. It is probably not to be used as often as space analysis, but the information provided is very precious. Mesh refinement, numerical boundary schemes or the effect of source terms can be studied with this tool. It is thus a good counterpart of space analysis. Moreover, it has the benefit to clearly introduce the notion of characteristic roots and the possible existence of several waves for a given pulsation.

6. Conclusion

This document has presented both space and time spectral analyses to study numerical methods in the frame of the convection-dominated phenomena. As in the works of Vichnevetsky [11] and Trefethen [9], wave propagation definitions, such as the dispersion relation and the group velocity, are introduced and are essential properties to assess a difference scheme. Space analysis is quite classical, as it is closely linked the Von Neumann analysis. The numerical examples given in section 5 have shown its benefits, especially the use of group velocity to predict the propagation of a wavepacket.

On the other hand, time analysis, introduced by Vichnevetsky and Bowles [11], is rare and essential yet. Indeed, when the space Fourier transform cannot be used, it is an available and powerful tool. As an example, it enables to determine the group velocities of numerical boundary schemes and thus the spurious reflections at the boundaries [11]. Another strong argument for time analysis is that Trefethen [9] has shown the close link between this work due to Vichnevetsky and Bowles [11] (which gives a physical point of view) and the powerful but complicated *G-K-S (Gustafsson-Kreiss-Sundström) theory* [3] (which is mathematically sound).

Any linear numerical scheme should be first evaluated with, at least, one of these two tools, before being used to solve more complicated problems (several dimensions, non-linear systems...). In no way, spectral analyses will give fully *sufficient* information about stability or errors for those difficult problems. Nevertheless, having good spectral qualities and robustness when solving Eq. (1.1) remains a *necessity*, if the scheme is used to solve accurately, for instance, the convective terms of the Navier-Stokes equations.

Those tools will allow to assess the performances of other schemes, like the Taylor-Galerkin family, in a document to come, and then generalized to realize multi-dimensional domains.

A. Document Revision History

This appendix summarises the main revisions of the present report.

Version	**Date**	**Comment**
1.0	2011-01	Initial report
1.1	2026-08	Few corrections

Table 2: Document revision history

References

[1] Bailly, C., Bogey, C., 2004. A family of low dispersive and low dissipative explicit schemes for noise computation. Journal of Computational Physics 194, 194–214.

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